ZARI.ai EfficientNetV2-B2, EDL & SCRC System Report

Model Selection Rationale, Total Parameters, Dirichlet EDL Uncertainty & SCRC Safety Gates

1. Why EfficientNetV2-B2 Was Selected

EfficientNetV2-B2 was chosen as the backbone architecture for ZARI.ai's 2-stage hierarchical vision pipeline over heavy Vision Transformers (Swin-Base, ViT-B/16) and ResNet baselines after rigorous empirical benchmark evaluations.

Key Architecture Advantages

- Fused-MBConv Layers: Replaces early 3x3 depthwise convolutions with standard 3x3 convolutions in lower stages, boosting training GPU throughput by 3.1x.
- Parametric Efficiency: Achieves 99.48% Crop Routing Accuracy with ~7.77 Million parameters per model (11x smaller than Swin-Base 88M).
- Progressive Learning: Jointly scales regularization and image resolution (384x384) during training to prevent over-fitting on plant leaf datasets.
- Ultra-Fast GPU Latency: ~3.2 ms CUDA latency per image tensor, enabling real-time edge execution.

2. Total System Parameters & Empirical Checkpoints Registry

ZARI.ai utilizes a 2-Stage Hierarchical suite of 4 distinct EfficientNetV2-B2 PyTorch models. The table below lists exact empirical state dict parameter counts and checkpoint file sizes on disk:

Model Component	Architecture	Output Head	Exact Params	Checkpoint Path	Metric / Accuracy
Model A (Router)	EfficientNetV2-B2	3 Crop Classes	7,772,858	best_model_a_efficientnetv2_b2.	99.48% Accuracy
Model B (Tomato)	EDLEfficientNetB2	13 Diseases	7,786,948	best_model_b_tomato.pth	0.9787 Macro F1
Model B (Potato)	EDLEfficientNetB2	3 Diseases	7,772,858	best_model_b_potato.pth	0.9718 Macro F1
Model B (Pepper)	EDLEfficientNetB2	6 Diseases	7,777,085	best_model_b_pepper.pth	0.9963 Macro F1
TOTAL SYSTEM	4-Model Suite	22 Classes Total	31,109,749	355.73 MB Total Checkpoints	99.1% Reliability

3. Evidential Deep Learning (EDL) Formulation

Standard neural networks use Softmax to output point probabilities, which causes overconfidence on out-of-distribution (OOD) crops (e.g. Corn, Wheat). EDL replaces Softmax with Subjective Logic and Dirichlet distributions:

1. Evidence Softplus: e_k = Softplus(z_k) = ln(1 + exp(z_k))
2. Dirichlet Alpha Params: alpha_k = e_k + 1.0
3. Total Dirichlet Intensity: S = sum_{k=1}^K alpha_k
4. Evidential Uncertainty: u = K / S = K / sum(e_k + 1)
5. Class Probabilities: p_k = alpha_k / S

For an OOD crop image (e.g., Corn or Wheat), the model accumulates zero evidence (e_k -> 0), so S -> K, resulting in Evidential Uncertainty u -> 1.0 (High Uncertainty).

4. Selective Classification Risk Control (SCRC)

SCRC enforces strict statistical risk control to limit the false acceptance rate of misclassified or OOD images to <= 5.0%:

Calibrated Safety Threshold: tau_SCRC = 0.3175 (from scrc_threshold.json)
Model A Confidence Gate: p_crop >= 0.80 (Rejects non-target crops like Corn/Wheat)
Model B Confidence Gate: p_disease >= 0.70
SCRC Action Rule: Accept if (u <= 0.3175 AND p_crop >= 0.80 AND p_disease >= 0.70) else REJECT

5. Production Input/Output & Latency Specification

Input Image Shape:	RGB Image File (JPEG/PNG/WebP), resized to (384, 384, 3)
Normalization Stats:	Mean = [0.485, 0.456, 0.406], Std = [0.229, 0.224, 0.225]
Vision GPU Latency:	3.24 ms CUDA inference time
Offline End-to-End Latency:	9.40 ms (Vision + ChromaDB RAG + Offline Synthesis)
Online LLM Latency:	389 ms (Vision + ChromaDB RAG + Groq Llama 3.1 8B API)
Output Payload:	status, disease_class, confidence, uncertainty, scrc_threshold, advisory